Project Dashboard - Comprehensive Project Tracking System

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1. Project Basic Information

Project Name:	Plant Monitoring Sensor Network		
Client Name:	John bambery		
Start Date:	25-Oct-25	**Expected End Date:**	28-Feb-26
Pricing Model:	**Fixed Price** ☐ **Hourly Rate** ☑		
Hourly Rate Per Hour ($):	$20.87	**Total Project Budget:**	5000$
Payment by Milestone:	☐		

2. Project Requirements

Project Requirements:

The project involves designing and building an MVP prototype of a multi-modal plant monitoring system. It includes two battery-powered, weatherproof sensor nodes integrating environmental sensors (soil moisture, temperature, humidity, light, pH, atmospheric pressure) and plant communication sensors (bioelectric potential, VOCs, IR leaf temperature, optional ultrasonic). Each node must support data collection, logging, and LoRa transmission, with firmware for sensor calibration, power management, and basic visualization.

3. Project Milestones

Start Date:

	Task Performed	Hours Spent	Description	Status
Milestone 1 – System Architecture, Hardware Development & Sensor Validation	1. System Architecture Design • Designed the complete sensor network architecture. • Defined sensor roles, data flow, and communication protocol. • Finalized I2C-based inter-module communication.	12	Planning & Initial Testing	Complete
	2. Sensor Research & Finalization • Researched and finalized 13 plant and environmental sensors. • Compared suppliers including Adafruit and DFRobot. • Coordinated with the client for technical validation.	26		Complete
	3. Bill of Materials & Procurement • Prepared BOM covering all sensors and components. • Verified specifications, availability, and datasheets. • Coordinated payment and procurement from multiple vendors. • Received and verified all components.	8		Complete
	4. Individual Sensor Testing & Firmware Development • Built breadboard prototypes and developed offline programs for each sensor. • Verified sensor data outputs and stability. • Resolved interfacing issues using datasheets and supplier documentation. • Confirmed reliability of individual sensors.	10		Complete
	5. Mechanical Assembly & Enclosure Customization • Designed and customized enclosures. • Installed connectors, push buttons, and wiring pathways. • Performed soldering, wire routing, and structural adjustments. • Created detailed wiring diagrams.	8		Complete
	6. Multi-Module Integration & I2C Network Debugging • Distributed 13 sensors across multiple modules. • Connected modules via I2C bus and resolved communication issues. • Troubleshot packet loss and inter-module data errors. • Validated stable multi-module data transmission.	12		Complete
	Milestone 1 Hours :	76		

	Task Performed	Hours Spent	Description	Status
Milestone 2-Power Optimization, Backend Development & Full System	1. Multi-Module Wiring Stabilization • Fixed wiring inconsistencies between sensor modules. • Improved I2C signal reliability over long cable runs.	4	Power & Connectivity	Complete
	2. Power Optimization & Energy Management • Identified excessive battery drain due to high duty cycle operation. • Implemented deep sleep in firmware with optimized wake intervals. • Resolved data interruptions during deep sleep. • Validated stable battery-powered operation.	14		Complete
	3. Backend Architecture & API Development • Designed backend system and using FastAPI. • Structured database schema for sensor data storage. • Implemented JSON-based data formatting. • Fixed data delivery and storage bugs.	24		Complete

Integration	4. Database & Data Integrity Management • Ensured proper logging of sensor values, device status, and timestamps. • Verified data persistence and consistency.	7	Connectivity (Backend and Frontend)	Complete
	5. Frontend Web Dashboard Development • Designed web dashboard for real-time monitoring. • Integrated frontend with backend API and database for seamless data display. • Ensured accurate visualization of sensor data.	22		Complete
	6. Full System Integration Testing • Integrated optimized firmware with backend and frontend. • Resolved deep sleep communication issues. • Verified sensor readings and data accuracy under real environmental conditions. • Conducted end-to-end system testing.	12		Complete
	Milestone 2 Hours :	83		
Milestone 3- Final Assembly, Protection, Documentation & Delivery	**Task Performed**	**Hours Spent**	**Description**	**Status**
	1. Final Hardware Assembly • Completed structured wiring and module packaging. • Secured connections for field stability.	6	Final Assembly & Delivery	Complete
	2. Waterproofing & Environmental Protection • Applied silicone sealing and protective enclosures. • Ensured durability for outdoor deployment. • Labeling the ports and the Modules.	3		Complete
	3. Final Validation & Field Testing • Re-tested complete system after sealing. • Verified sensor accuracy, connectivity, and battery performance.	5		Complete
	4. Packaging & Delivery Preparation • Prepared protective packaging for safe transportation. • Conducted final inspection of the complete system.	2		Complete
	5. Documentation Preparation • Created schematic diagrams and final BOM. • Developed system architecture documentation and user manual. • Prepared setup and operational guidelines for the client.	8		Complete
	6. Project Delivery • Delivered fully assembled, tested, and documented plant-based sensor network system via Fedex.	1		Complete
	Milestone 3 Hours :	25		

4. Key Deliverables		Additional Requirements	Project End Goal and Approval Criteria
Sensor Selection	☑	**Hardware**	All hardware components functioning to specification
Firmware Development	☑	LoRa communication or WIFI	Software meets All requirements with no critical bugs
Schematic Drawing	☐	Assembled sensor nodes	Client signs off on final deliverables
PCB Layout	☐	Power requirements	
Gerber Files	☐	Solar battery system	
Bill of Materials (BOM)	☐	**Software**	**Design**
PCB Fabrication Ordering	☐	Firmware sensor integration	IP67 weatherproof enclosure
PCB Assembly (soldering components)	☑	Sensor calibration option	Outdoor mounting system and Compact modular design
Board Testing	☐	LED status indicators	Corrosion-resistant materials
3D Enclosure Design	☐		

6. Additional Expenses :

Item / Description	Category	Notes	Cost per Unit ($)	Units	Total ($)
Components, Hardware and Sensors	Components (Adafruit, Dfrobot Digilog, Mouser electronics)	Separate bill for sensors and purchased materials for the project, valued at 2,762 EUR, equivalent to 3,279.78 USD at the USD-to-EUR exchange rate of 1 EUR = 1.19 USD.	$3,286.00	1	$3,286.00
Delivery to Client	Logistics	I haven't received the invoice yet for FedEx International Economy (Tracking 888703177458). I will share it once the package is delivered and my card has been charged. This is what was shown when I placed the order for delivery.	$201.80	1	$201.80
Received Amount	**Amount details as mentioned in the Payment History section**		**$3,465.12**	**Remaining Balance:**	**$22.68**

7. Received Payments History

Purpose	Date	Discription	Amount ($)	Paid ($)
Fiverr Order	8-Nov-25	Recived order on Fiverr	$500.00	$500.00
Project Components and Sensors	13-Nov-25	Received 2,985 EUR for component purchasing, equivalent to 3,465.12 USD at the USD-to-EUR exchange rate of 0.86 on that day.	$3,465.12	$3,465.12
Engineering and Manufacturing costs	4-Dec-25	Received 1,249.51 EUR toward project engineering and manufacturing costs, equivalent to 1,452.92 USD at the USD-to-EUR exchange rate of 0.86 on the date received.	$1,452.90	$1,452.90
Total Amount:		**0**	**$5,418.02**	**$5,418.02**

8. Remaining Payments

Hourly Payments

Task / Milestone	Rate ($/hr)	Hours Worked	Amount ($)	Paid ($)	Remaining ($)
Milestone 1	$20.87	76	$1,586.12	$500.00	$1,086.12
Milestone 2	$20.87	83	$1,732.21	$1,452.90	$279.31
Milestone 3	$20.87	25	$521.75		$521.75
Additional Expenses	Nil	Remaining payable for components and delivery.	$22.68		$22.68
Total Amount:		**184**	**$3,862.76**	**$1,952.90**	**$1,909.86**

Thank you for choosing to work with me. I truly appreciate your trust, and I did my best to bring your idea to life with precision, dedication, and transparency. With this stage complete, we will continue the Lora integration and work on one additional node. This spreadsheet provides a clear overview of project scope, progress, and costs, ensuring full visibility at every step.

Please Note:
- The document covers total project costs, including components, payments received, engineering and Mechanical work, prototype development, and delivery charges.
- The hardware is already on the way and should arrive next week.
- The remaining amount of $1,909 can be postponed until next month; a separate Upwork order can be created to process this payment or we will decide.

Warm regards,
M. Zaeem
Electronics & Embedded Systems Engineer